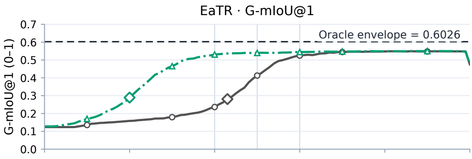
9-backbone exploratory subset · GMR baseline vs GEQ+aux

Fixed Val-selected checkpoints → Standard Test (1,036 queries)

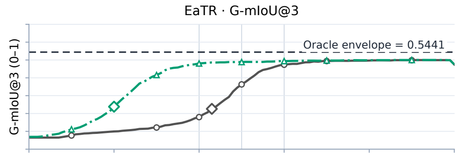
Diamonds: Val ≥86% retention operating points (0.01 grid); ΔG1/G3 refer to these points.

Exploratory subset chosen after Test scan; excluded: Moment-DETR and LD-DETR.

EaTR · G-mIoU@1



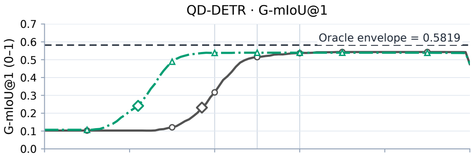
EaTR · G-mIoU@3



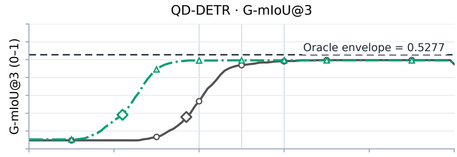
GMR baseline | mAP 16.37% | τ86 0.43 — GEQ + aux | mAP 17.16% | τ86 0.20

ΔmAP: +0.79 pp — ΔG1 at τ86: +0.85 pp — ΔG3 at τ86: +1.21 pp

QD-DETR · G-mIoU@1



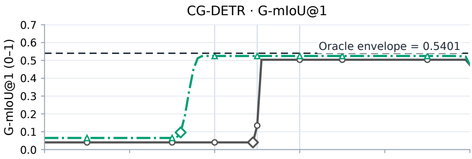
QD-DETR · G-mIoU@3



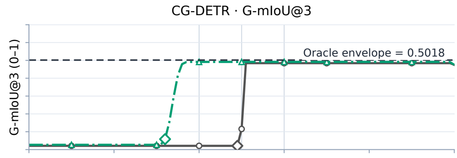
GMR baseline | mAP 11.56% | τ86 0.37 — GEQ + aux | mAP 13.05% | τ86 0.22

ΔmAP: +1.49 pp — ΔG1 at τ86: +0.98 pp — ΔG3 at τ86: +1.25 pp

CG-DETR · G-mIoU@1



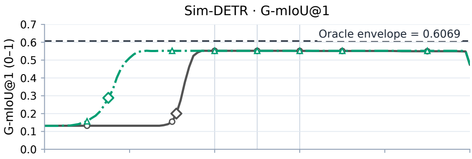
CG-DETR · G-mIoU@3



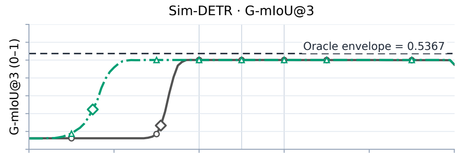
GMR baseline | mAP 6.37% | τ86 0.49 — GEQ + aux | mAP 6.91% | τ86 0.32

ΔmAP: +0.54 pp — ΔG1 at τ86: +5.60 pp — ΔG3 at τ86: +3.75 pp

Sim-DETR · G-mIoU@1



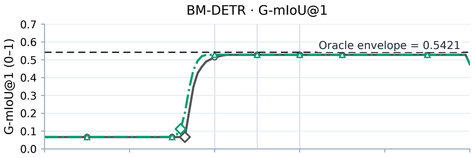
Sim-DETR · G-mIoU@3



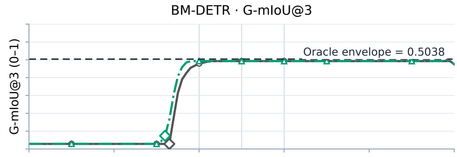
GMR baseline | mAP 20.29% | τ86 0.31 — GEQ + aux | mAP 21.00% | τ86 0.15

ΔmAP: +0.71 pp — ΔG1 at τ86: +8.81 pp — ΔG3 at τ86: +9.01 pp

BM-DETR · G-mIoU@1



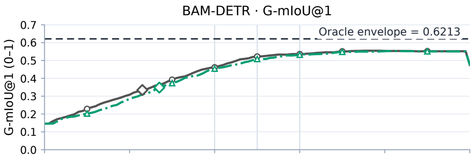
BM-DETR · G-mIoU@3



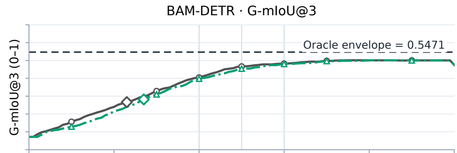
GMR baseline | mAP 4.11% | τ86 0.33 — GEQ + aux | mAP 4.20% | τ86 0.32

ΔmAP: +0.09 pp — ΔG1 at τ86: +4.51 pp — ΔG3 at τ86: +4.57 pp

BAM-DETR · G-mIoU@1



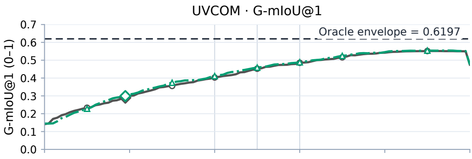
BAM-DETR · G-mIoU@3



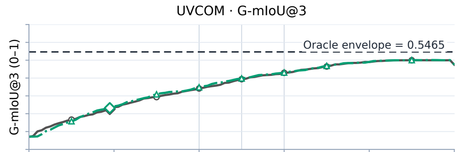
GMR baseline | mAP 22.97% | τ86 0.23 — GEQ + aux | mAP 23.64% | τ86 0.27

ΔmAP: +0.67 pp — ΔG1 at τ86: +1.38 pp — ΔG3 at τ86: +1.52 pp

UVCOM · G-mIoU@1



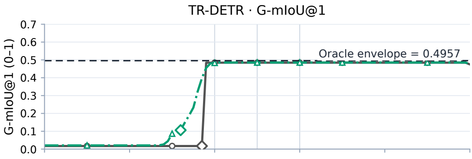
UVCOM · G-mIoU@3



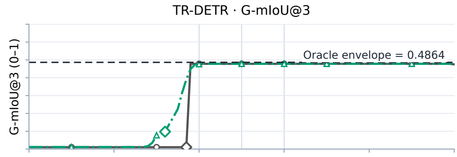
GMR baseline | mAP 22.21% | τ86 0.19 — GEQ + aux | mAP 21.66% | τ86 0.19

ΔmAP: -0.55 pp — ΔG1 at τ86: +0.99 pp — ΔG3 at τ86: +0.61 pp

TR-DETR · G-mIoU@1



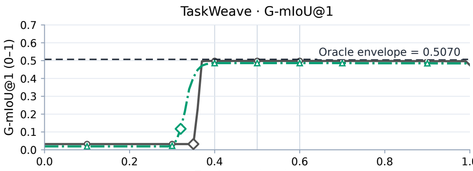
TR-DETR · G-mIoU@3



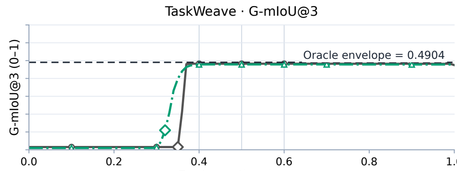
GMR baseline | mAP 2.58% | τ86 0.37 — GEQ + aux | mAP 2.81% | τ86 0.32

ΔmAP: +0.23 pp — ΔG1 at τ86: +8.90 pp — ΔG3 at τ86: +8.70 pp

TaskWeave · G-mIoU@1



TaskWeave · G-mIoU@3



GMR baseline | mAP 3.79% | τ86 0.35 — GEQ + aux | mAP 3.40% | τ86 0.32

ΔmAP: -0.39 pp — ΔG1 at τ86: +8.54 pp — ΔG3 at τ86: +9.40 pp

Oracle envelope: max of the two methods with GT presence and each method's fixed candidates; analysis only.

Curves: 101 fixed thresholds, 0–1 score units. τ86 uses Standard Val only; the same threshold is used on both Tests.

mAP is ungated; Δ = GEQ+aux − matching auxiliary-trained GMR baseline. Subset selected on Test; all11 results retained.

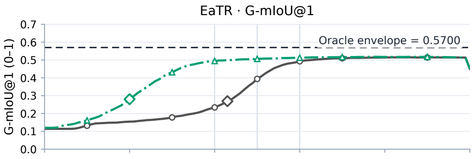
9-backbone exploratory subset · GMR baseline vs GEQ+aux

Fixed Val-selected checkpoints → Full-Decodable Test (2,970 / 2,986 queries)

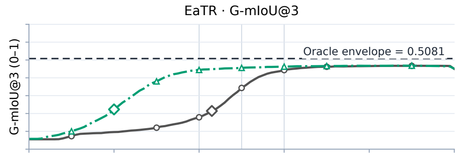
Diamonds: Val ≥86% retention operating points (0.01 grid); ΔG1/G3 refer to these points.

Exploratory subset chosen after Test scan; excluded: Moment-DETR and LD-DETR.

EaTR · G-mIoU@1



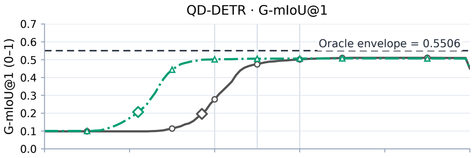
EaTR · G-mIoU@3



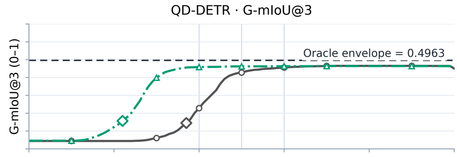
GMR baseline | mAP 13.77% | τ86 0.43 — GEQ + aux | mAP 14.68% | τ86 0.20

ΔmAP: +0.91 pp · ΔG1 at τ86: +1.10 pp · ΔG3 at τ86: +0.87 pp

QD-DETR · G-mIoU@1



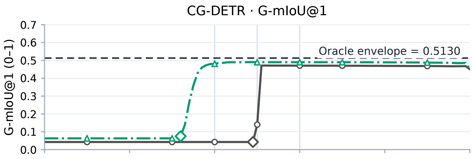
QD-DETR · G-mIoU@3



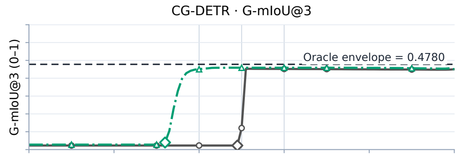
GMR baseline | mAP 10.50% | τ86 0.37 — GEQ + aux | mAP 11.22% | τ86 0.22

ΔmAP: +0.72 pp · ΔG1 at τ86: +1.05 pp · ΔG3 at τ86: +1.17 pp

CG-DETR · G-mIoU@1



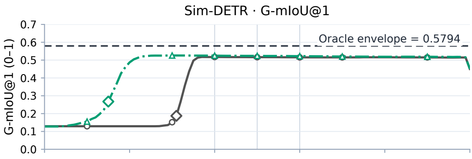
CG-DETR · G-mIoU@3



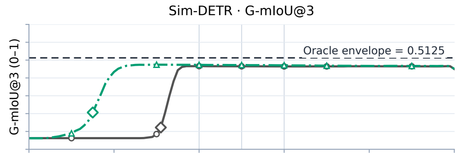
GMR baseline | mAP 5.82% | τ86 0.49 — GEQ + aux | mAP 6.19% | τ86 0.32

ΔmAP: +0.37 pp · ΔG1 at τ86: +3.21 pp · ΔG3 at τ86: +1.69 pp

Sim-DETR · G-mIoU@1



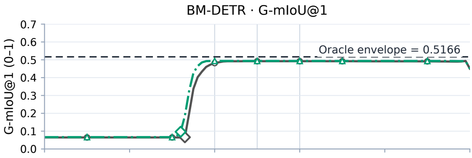
Sim-DETR · G-mIoU@3



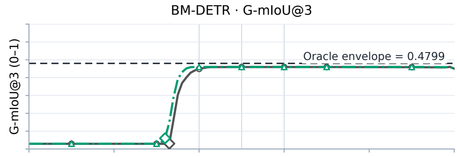
GMR baseline | mAP 19.60% | τ86 0.31 — GEQ + aux | mAP 19.46% | τ86 0.15

ΔmAP: -0.14 pp · ΔG1 at τ86: +8.02 pp · ΔG3 at τ86: +8.41 pp

BM-DETR · G-mIoU@1



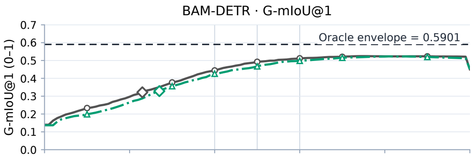
BM-DETR · G-mIoU@3



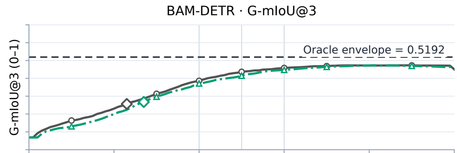
GMR baseline | mAP 3.87% | τ86 0.33 — GEQ + aux | mAP 3.81% | τ86 0.32

ΔmAP: -0.06 pp · ΔG1 at τ86: +3.07 pp · ΔG3 at τ86: +2.98 pp

BAM-DETR · G-mIoU@1



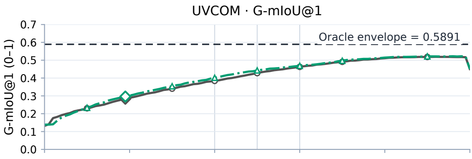
BAM-DETR · G-mIoU@3



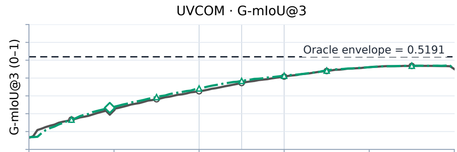
GMR baseline | mAP 20.33% | τ86 0.23 — GEQ + aux | mAP 20.59% | τ86 0.27

ΔmAP: +0.26 pp · ΔG1 at τ86: +0.66 pp · ΔG3 at τ86: +1.12 pp

UVCOM · G-mIoU@1



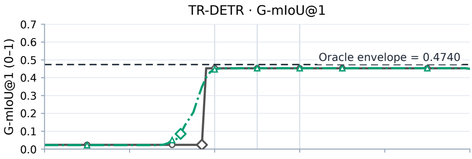
UVCOM · G-mIoU@3



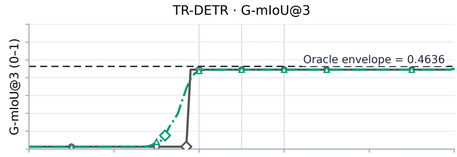
GMR baseline | mAP 18.46% | τ86 0.19 — GEQ + aux | mAP 19.21% | τ86 0.19

ΔmAP: +0.75 pp · ΔG1 at τ86: +1.28 pp · ΔG3 at τ86: +1.12 pp

TR-DETR · G-mIoU@1



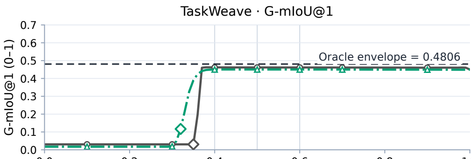
TR-DETR · G-mIoU@3



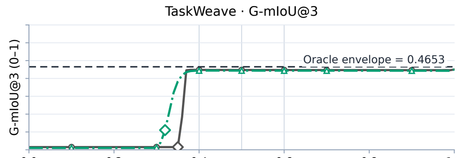
GMR baseline | mAP 2.85% | τ86 0.37 — GEQ + aux | mAP 2.84% | τ86 0.32

ΔmAP: -0.01 pp · ΔG1 at τ86: +6.16 pp · ΔG3 at τ86: +6.27 pp

TaskWeave · G-mIoU@1



TaskWeave · G-mIoU@3



GMR baseline | mAP 3.70% | τ86 0.35 — GEQ + aux | mAP 3.31% | τ86 0.32

ΔmAP: -0.39 pp · ΔG1 at τ86: +8.63 pp · ΔG3 at τ86: +9.52 pp

Oracle envelope: max of the two methods with GT presence and each method's fixed candidates; analysis only.

Curves: 101 fixed thresholds, 0–1 score units. τ86 uses Standard Val only; the same threshold is used on both Tests.

mAP is ungated; Δ = GEQ+aux − matching auxiliary-trained GMR baseline. Subset selected on Test; all11 results retained.